



## CSA0501-DATABASE MANAGEMENT SYSTEMS

### LAB EXPERIMENTS

DATE: 2/08/2026

#### EXPERIMENT 14 -- TCL COMMANDS – COMMIT, SAVEPOINT, ROLLBACK

##### AIM:

To learn how to use various TCL commands Commit, Savepoint and Rollback SQL commands

##### PROCEDURE:

1. Select the database containing the student\_cgpa table.
2. Display the existing records using SELECT.
3. Start a transaction using START TRANSACTION.
4. Perform an UPDATE operation on a student record.
5. Create **SAVEPOINT A**.
6. Perform an INSERT operation.
7. Create **SAVEPOINT B**.
8. Perform another INSERT operation.
9. Create **SAVEPOINT C**.
10. Use ROLLBACK TO to return to a particular savepoint.
11. Use SELECT to verify the changes.
12. Finally, use COMMIT to permanently save the required changes.

##### PROGRAM:

###### 1. CREATE TABLE

```
CREATE TABLE student_cgpa (  
    reg_no VARCHAR(15) PRIMARY KEY,  
    student_name VARCHAR(50) NOT NULL,  
    department VARCHAR(20),  
    no_of_subs_pass INT,  
    no_of_subs_fail INT,  
    cgpa DECIMAL(3,2)  
);
```

###### Then Insert Values:

```
INSERT INTO student_cgpa  
(reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, cgpa)
```

VALUES

```
('192411123','Deepthi','CSE',22,2,8.71),
('192512121','Varun','IT',27,1,9.23),
('192312117','Aadhya','ECE',25,3,8.97),
('192411227','Dimple','CSE',31,0,9.37),
('192820896','Mohan','EEE',18,9,6.48),
('192411122','Yogitha','CSE',31,1,9.01),
('192516871','Sadhya','IT',25,5,7.22),
('192111101','Rahul','MECH',30,12,5.14),
('192411010','Alekhyia','CSE',35,2,9.01),
('192511135','Rahul','CIVIL',37,0,9.72);
```

```
SELECT * FROM student_cgpa;
```

**OUTPUT:**

The screenshot shows a database management tool interface. At the top, there's a 'Result Grid' with columns: reg\_no, student\_name, department, no\_of\_subs\_pass, no\_of\_subs\_fail, and cgpa. The data is as follows:

| reg_no    | student_name | department | no_of_subs_pass | no_of_subs_fail | cgpa |
|-----------|--------------|------------|-----------------|-----------------|------|
| 192111101 | Rahul        | MECH       | 30              | 12              | 5.14 |
| 192312117 | Aadhya       | ECE        | 25              | 3               | 8.97 |
| 192411010 | Alekhyia     | CSE        | 35              | 2               | 9.01 |
| 192411122 | Yogitha      | CSE        | 31              | 1               | 9.01 |
| 192411123 | Deepthi      | CSE        | 22              | 2               | 8.71 |
| 192411227 | Dimple       | CSE        | 31              | 0               | 9.37 |
| 192511135 | Rahul        | CIVIL      | 37              | 0               | 9.72 |
| 192512121 | Varun        | IT         | 27              | 1               | 9.23 |
| 192516871 | Sadhya       | IT         | 25              | 5               | 7.22 |
| 192820896 | Mohan        | EEE        | 18              | 9               | 6.48 |

Below the table, there's an 'Action Output' section showing the execution of SQL commands:

| # | Time     | Action                                                                                                    | Message                                                  | Duration / Fetch      |
|---|----------|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------|-----------------------|
| 1 | 22:41:47 | CREATE TABLE student_cgpa ( reg_no VARCHAR(15) PRIMARY KEY, student_name VARCHAR(50) NO...                | 0 row(s) affected                                        | 0.031 sec             |
| 2 | 22:44:31 | INSERT INTO student_cgpa (reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, cgpa) VA... | 10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0 | 0.000 sec             |
| 3 | 22:46:18 | SELECT * FROM student_cgpa LIMIT 0, 1000                                                                  | 10 row(s) returned                                       | 0.000 sec / 0.000 sec |

## 1. COMMIT

```
START TRANSACTION;
```

```
UPDATE student_cgpa
SET cgpa = 8.80
WHERE reg_no = 192411123;
```

```
COMMIT;
SELECT * FROM student_cgpa
WHERE reg_no = 192411123;
```

**OUTPUT:**

The screenshot shows a database management tool interface. At the top, there's a 'Result Grid' with columns: reg\_no, student\_name, department, no\_of\_subs\_pass, no\_of\_subs\_fail, and cgpa. The data is as follows:

| reg_no    | student_name | department | no_of_subs_pass | no_of_subs_fail | cgpa |
|-----------|--------------|------------|-----------------|-----------------|------|
| 192411123 | Deepthi      | CSE        | 22              | 2               | 8.71 |
| NULL      | NULL         | NULL       | NULL            | NULL            | NULL |

## 2. SAVEPOINT

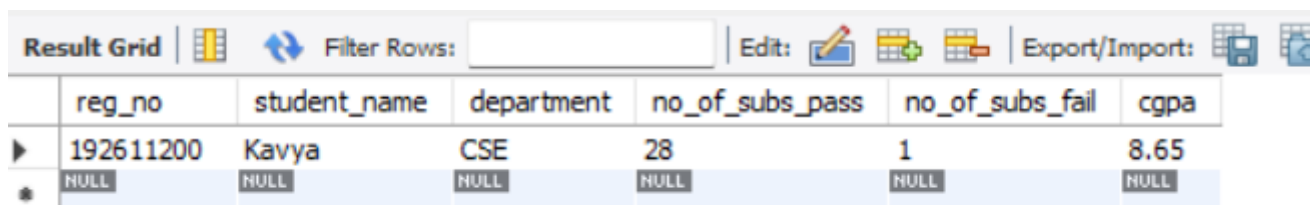
**Insert one student and create a savepoint.**

```
START TRANSACTION;
```

```
INSERT INTO student_cgpa  
(reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, cgpa)  
VALUES  
(192611200, 'Kavya', 'CSE', 28, 1, 8.65);
```

```
SAVEPOINT A;  
SELECT * FROM student_cgpa  
WHERE reg_no = 192611200;
```

### OUTPUT:



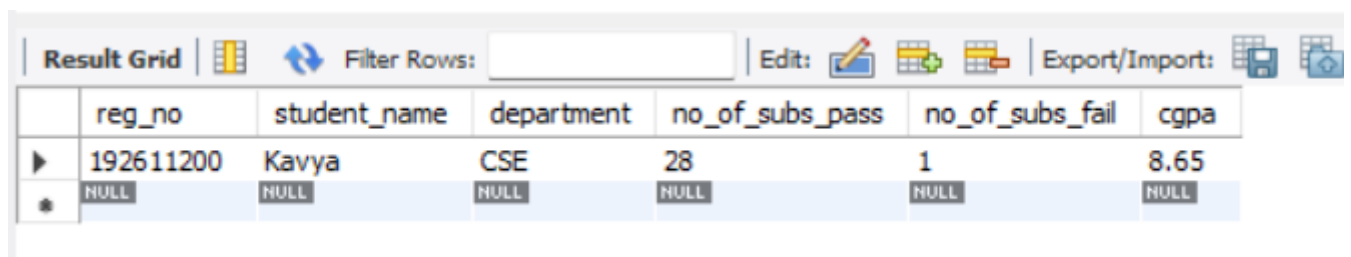
|   | reg_no    | student_name | department | no_of_subs_pass | no_of_subs_fail | cgpa |
|---|-----------|--------------|------------|-----------------|-----------------|------|
| ▶ | 192611200 | Kavya        | CSE        | 28              | 1               | 8.65 |
| ★ | NULL      | NULL         | NULL       | NULL            | NULL            | NULL |

## 3. ROLLBACK

**Rollback the insertion using the savepoint.**

```
ROLLBACK TO A;  
SELECT * FROM student_cgpa  
WHERE reg_no = 192611200;
```

### OUTPUT:



|   | reg_no    | student_name | department | no_of_subs_pass | no_of_subs_fail | cgpa |
|---|-----------|--------------|------------|-----------------|-----------------|------|
| ▶ | 192611200 | Kavya        | CSE        | 28              | 1               | 8.65 |
| ★ | NULL      | NULL         | NULL       | NULL            | NULL            | NULL |

## RESULT

Thus, the TCL commands COMMIT, SAVEPOINT, and ROLLBACK were successfully executed on the student\_cgpa table.